# 2020 INTRODUCTION TO PRICING

Bruin Actuarial Society

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# Agenda

- Rate Indications
  - On-levelling
  - Trending
  - Developing
- Classification Ratemaking

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# Rate Indications

- A **rate indication** indicates the rate change necessary on an aggregate basis to balance the **fundamental insurance equation**

$$\text{Premium} = \text{Losses} + \text{Loss Adjustment Expenses} + \text{Underwriting Expenses} + \text{Underwriting Profit}$$

  in the *prospective period*.
- We restate historic data at current levels to do this

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# Balancing the Fundamental Insurance Equation

- We will cover three main adjustments to historic premiums and losses:
  - On-levelling
  - Trending
  - Developing
- For simplicity, we will not cover adjustments to expenses or profit

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# On-Levelling Premium

- **On-levelling** refers to adjusting historic data at the current level.
  - When we on-level premium, we adjust for rate increases already taken
  - When we on-level losses, we adjust for changes in coverage levels, etc.
- The purpose of this is to avoid neglecting changes we have already taken, as this could skew our results.
- To understand this, let's see what happens if we do and don't on-level premium in the following example:

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# On-Levelling Premium: Example

| Date | Rate Change |
|---|---|
| 1/1/2018 | 0% |
| 1/1/2020 | +10% |
| 4/1/2020 | ??? |

- From 1/1/2018 to 12/31/2019, we charge $100 on average.
  - From 1/1/2020 to 3/31/2020, we charge $110 on average.
- Suppose the “true” rate for the future period is $120.
- If we do not on-level premium, looking at 2018-2019 data tells us we should increase rates by 120/100 – 1 = 20%.
  - However, we currently charge $110! Increasing by 20% will cause our rates to be $132, so we are overcharging!

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# On-Levelling Premium: Example

| Date | Rate Change |
|---|---|
| 1/1/2018 | 0% |
| 1/1/2020 | +10% |
| 4/1/2020 | ??? |

- We should adjust our 2018-2019 data for rate increases already taken:
  - Our $100 rate should be increased by 10% to $110.
- Now, we can use data from 1/1/2018 to 3/31/2020 on the same basis.
- (We can do the same process for losses.)

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# Trending Premium and Losses

- We **trend** premium and losses to their expected future values.
- These can differ from on-levelled premium and losses for any of the following reasons:
  - Composition of our book of business is changing
  - Inflation
  - Changing costs of claims
  - Etc.

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# Trending Losses: Example

| Year | % Car Drivers | Average On-Level Loss |
|---|---|---|
| 2016 | 50% | $110 |
| 2017 | 55% | $109 |
| 2018 | 60% | $108 |
| 2019 | 65% | $107 |

- Consider an auto insurance company that insures the following:
  - Car: Average on-level loss $100
  - Van: Average on-level loss $120
- We are not doing a good job retaining van drivers.
- We want to predict our average loss for 2020.

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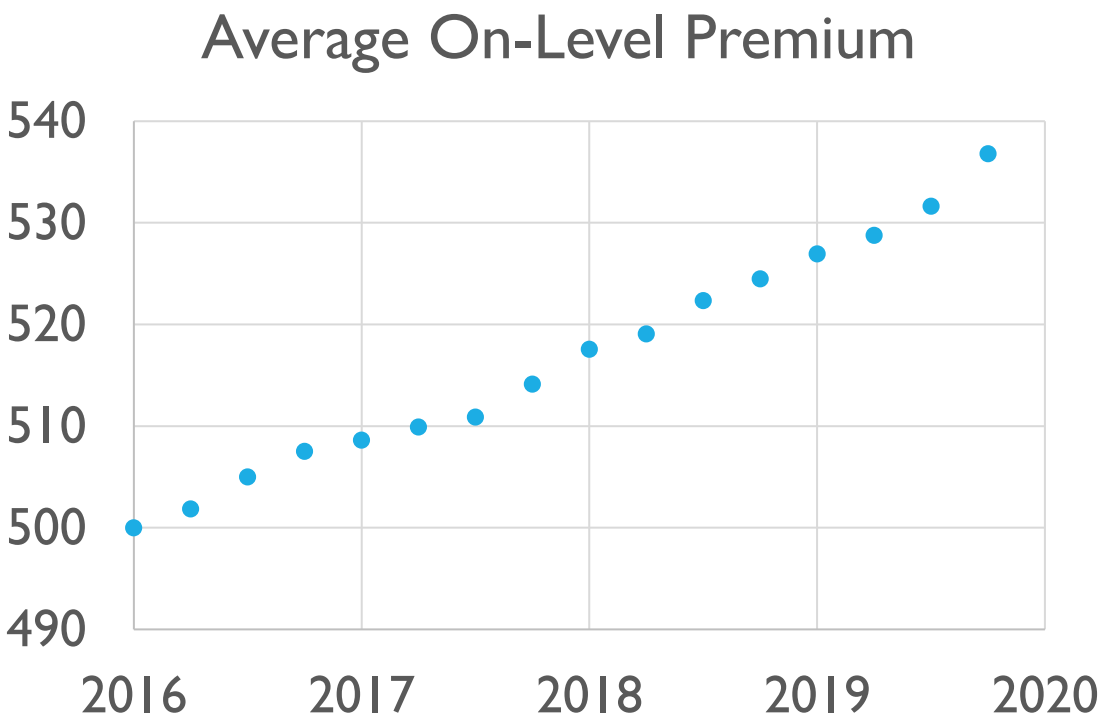
# Trending Losses: Example

| Year | Average On-Level Loss | % Change |
|---|---|---|
| 2016 | $110 | |
| 2017 | $109 | -0.9% |
| 2018 | $108 | -0.9% |
| 2019 | $107 | -0.9% |

- Our average losses are **decreasing** by about 0.9% per year!
- We should not use our average loss in 2016 of $110 without accounting for this 0.9% decrease per year.
  - Trending this data 4 years forward, we see that our average loss at expected 2020 levels is $106.

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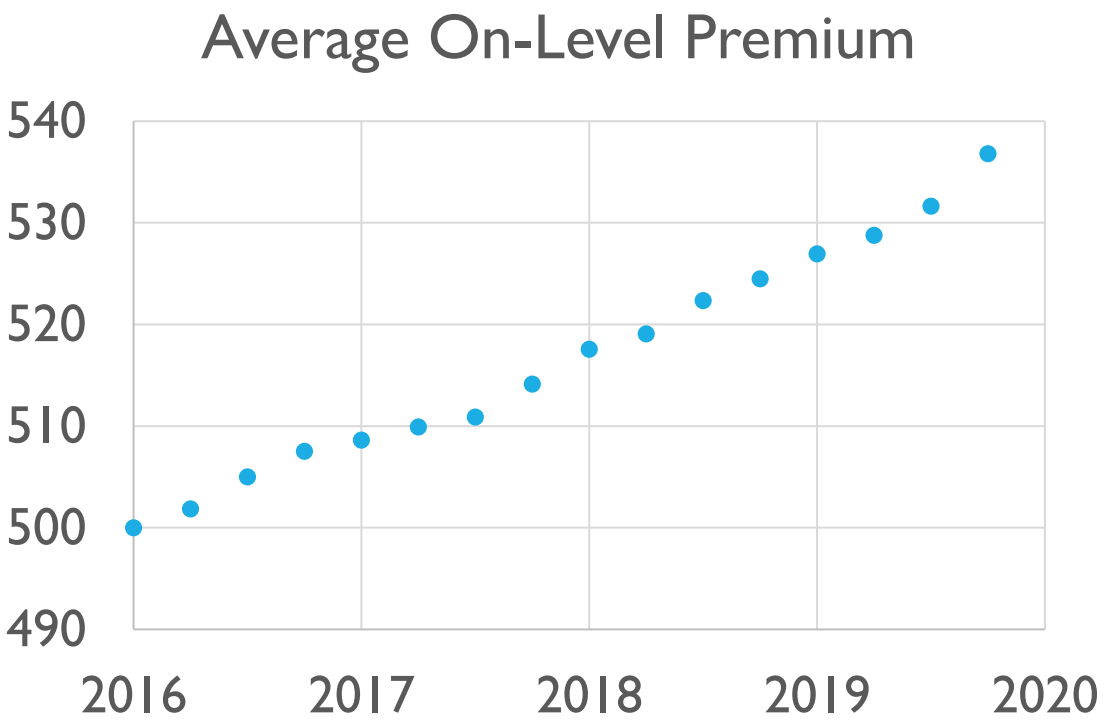
# Trending Premium: Example




- Let's consider a more complicated example.
- We will look at average on-level premium by **quarter**, and try to select a reasonable trend.

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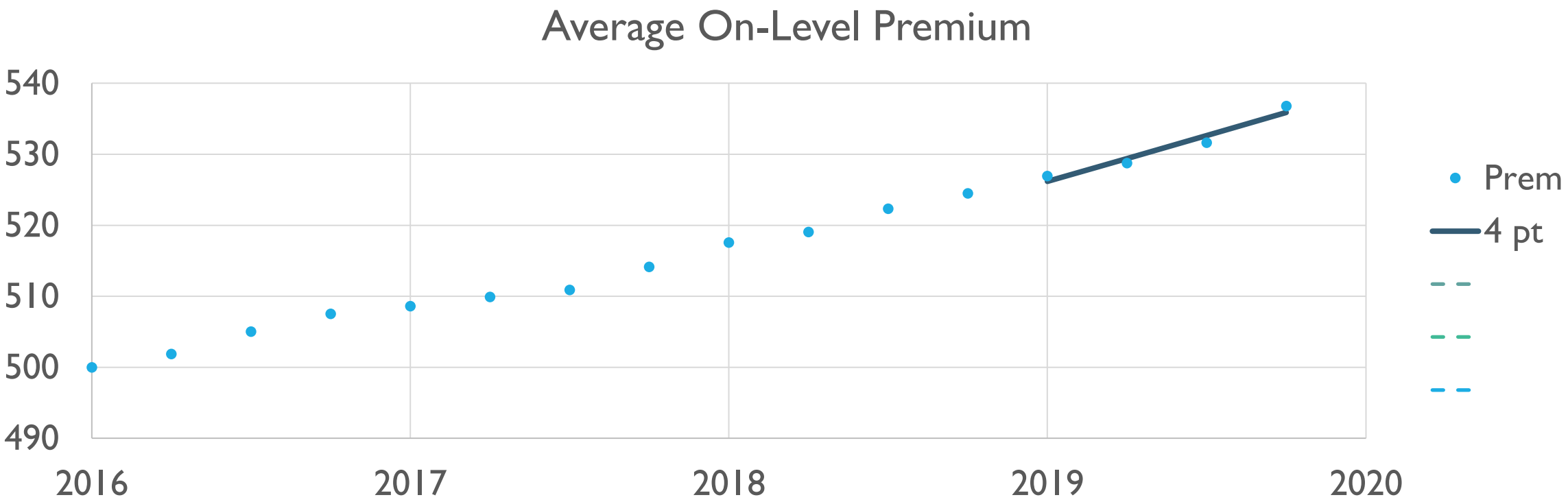
# Trending Premium: Example




- Typically, actuaries fit several exponential curves to this data; for example:
  - Trend using latest 4 points
  - Trend using latest 8 points
  - …

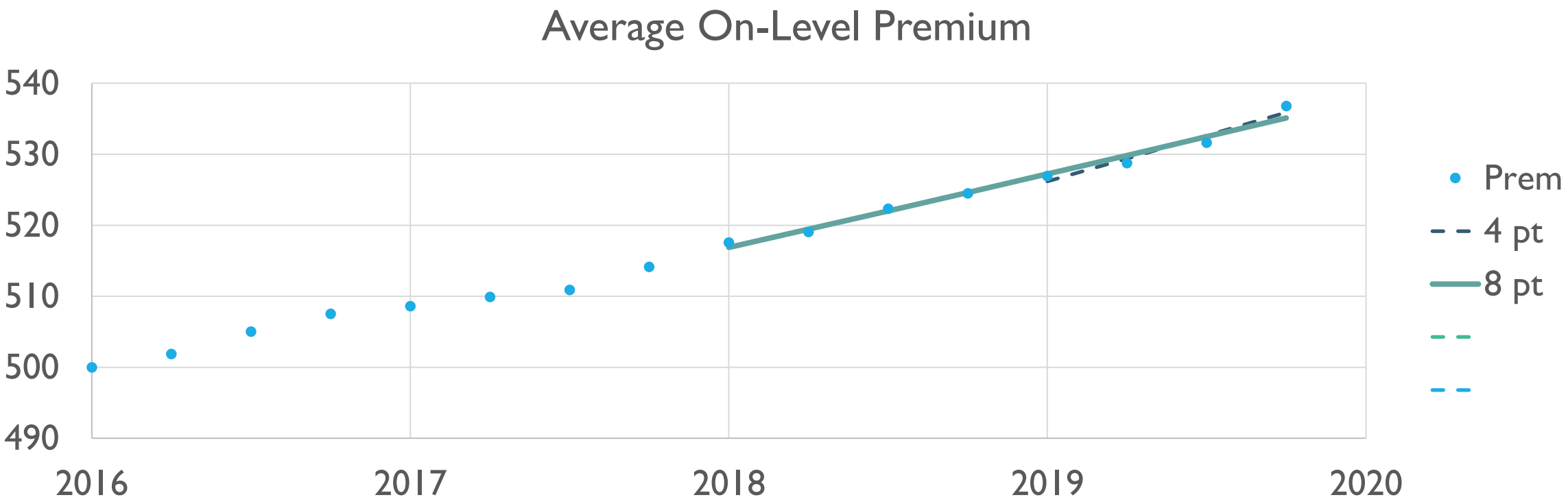
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# Trending Premium: Example



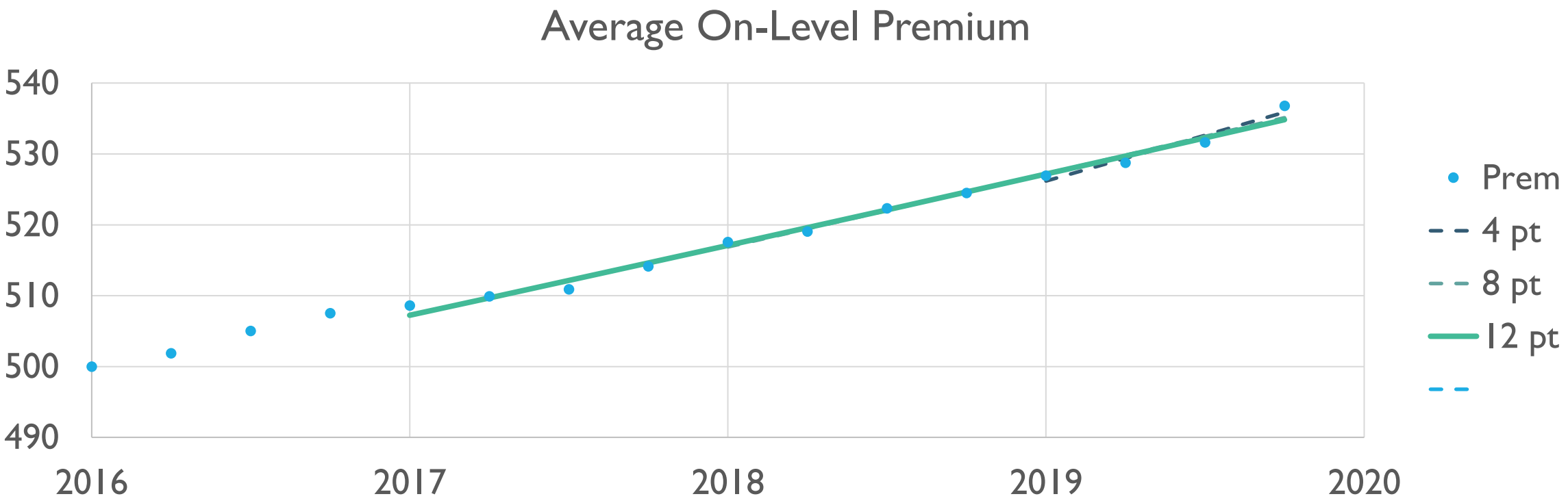

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# Trending Premium: Example



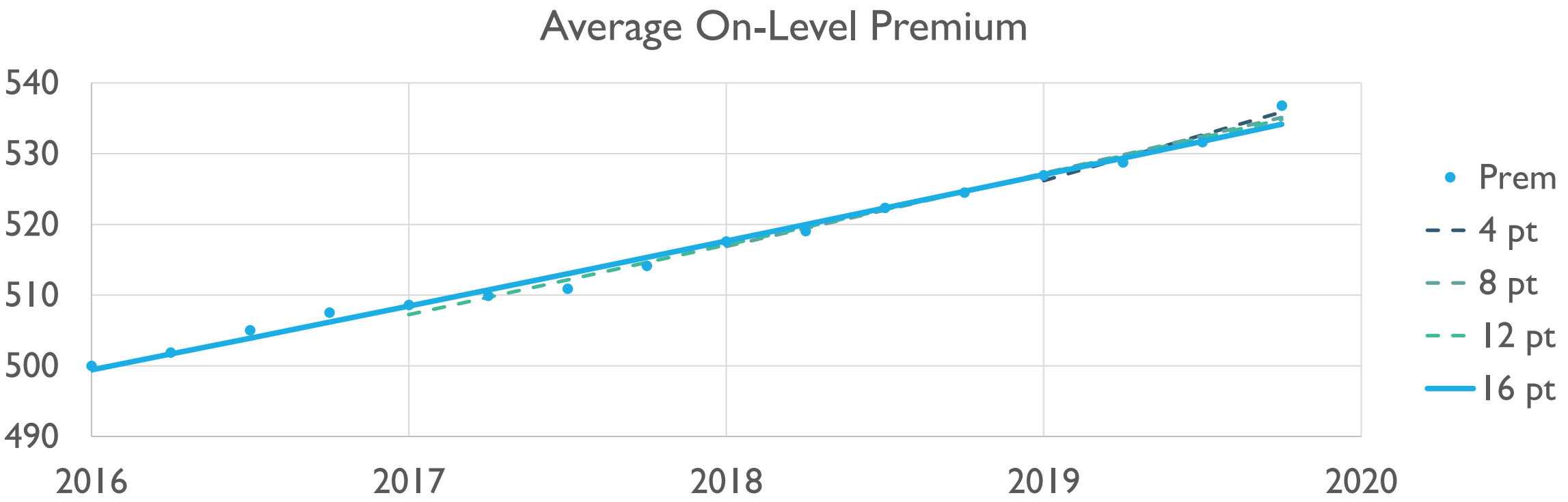

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# Trending Premium: Example



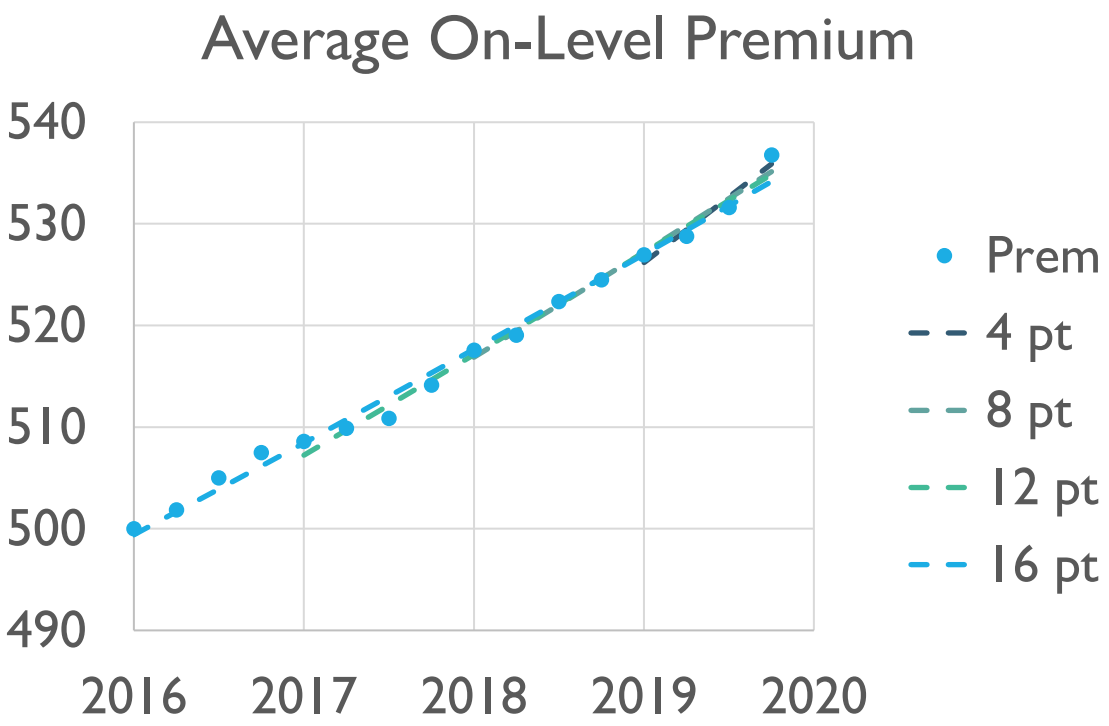

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# Trending Premium: Example




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# Trending Premium: Example




- 4-point trend: 2.5%
- 8-point trend: 2.0%
- 12-point trend: 1.9%
- 16-point trend: 1.8%
- Moral of the story: We can select different trends, and the most appropriate one is a judgmental choice.

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# Developing Losses

- We use **developed** losses in our equation, **not** reported or paid amounts.
- We can develop losses using the development method, the expected losses method, BF, etc. (*See Reserving Workshop!*)
- We must use ultimate losses, since the premium paid last year is meant to cover all covered losses from last year, even if we don't know about them yet!

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# Other Adjustments to Losses: Large Loss Load

- Often, we will remove the effect of large, catastrophic losses from analysis, then add them back in at the end.
- For example, consider the following hospital claims:

| Year | Average Hospital Claims |
|---|---|
| 2015 | $1,053 |
| 2016 | $1,105 |
| 2017 | $1,185 |
| 2018 | $1,250 |
| 2019 | $1,333 |
| 2020 | $1,900 |

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# Other Adjustments to Losses: Large Loss Load

- One might attribute the large increase in average claims in 2020 to COVID-19.
- Therefore, we might remove COVID-19 claims from calculation, then add a large loss load to account for such catastrophes.
  - We would include this in **every year**, not just 2020, as a sort of "expected large loss" amount.

| Year | Average Hospital Claims |
|---|---|
| 2015 | $1,053 |
| 2016 | $1,105 |
| 2017 | $1,185 |
| 2018 | $1,250 |
| 2019 | $1,333 |
| 2020 | $1,900 |

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# A Brief Commentary on Expenses and Profit

- **Allocated loss adjustment expenses** are typically analyzed together with losses (i.e. in triangles)
  - Recall that ALAE can be associated with a specific claim
- **Unallocated loss adjustment expenses** must be analyzed in aggregate, typically as a percentage of losses
  - ULAE cannot be linked to specific claims, so we can only look at payments made each calendar year

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# A Brief Commentary on Expenses and Profit

- **Underwriting expenses** are typically averaged in aggregate as well
  - Policy fees, taxes, commissions, and other expenses that scale with premium are generally estimated as *percentage of premium*
  - Marketing fees, salaries, and other expenses that are fixed with respect to premium are generally estimated as *dollars per exposure*
- **Underwriting profit** targets are set as a percentage of premium
  - Recall that other forms of profit (i.e. investment) are *not* part of the fundamental insurance equation, and are thus not considered when balancing the equation

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# Putting it All Together

- Once we have done all these steps, we will look at the following pieces of information by year:
  - Trended, on-level premium
  - Trended, on-level ultimate losses
  - Estimated prospective loss adjustment expenses
  - Estimated prospective underwriting expenses
  - Prospective underwriting profit provision
- We will use all these pieces to obtain our indicated rate change.

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# Putting it All Together: Example

| Year | Trended, On-Level Premium | Trended, On-Level Ultimate Losses + Prospective LAE + Prospective U/W Expense + U/W Profit Provision | Indicated Rate Change |
|---|---|---|---|
| 2016 | $15.0 M | $16.1 M | +7.3% |
| 2017 | $14.7 M | $15.3 M | +4.1% |
| 2018 | $16.2 M | $17.0 M | +4.9% |
| 2019 | $16.8 M | $17.7 M | +5.4% |
| **Total** | **$62.7 M** | **$66.1 M** | **+5.4%** |

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# Putting it All Together: Example

| Year | Adjusted Premium | Adjusted Losses + LAE + U/W Expense + U/W Profit | Indicated Rate Change |
|---|---|---|---|
| 2016 | $15.0 M | $16.1 M | +7.3% |
| 2017 | $14.7 M | $15.3 M | +4.1% |
| 2018 | $16.2 M | $17.0 M | +4.9% |
| 2019 | $16.8 M | $17.7 M | +5.4% |
| **Total** | **$62.7 M** | **$66.1 M** | **+5.4%** |

- We can add data from different rows because everything has been adjusted to the prospective level.
- We can also use a straight average or incorporate judgment into our choice of selected rate change.

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# Classification Ratemaking

- Recall that if we do not accurately reflect individual differences in rate, we will experience **adverse selection**. (*See Introduction Workshop!*)
- We will provide a brief example on how this is done on a univariate level.
- In practice, models such as GLMs consider all variables at once.

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# Classification Ratemaking: Example

| Marital Status | Current Factor | Indicated Rate |
|---|---|---|
| Single | 1.20 | $234 |
| **Married** | **1.00** | **$200** |
| Divorced | 1.10 | $222 |

- Consider an auto insurance company and the rating variable marital status.
- The **base class** is "married".
  - The base class is the class that, by definition, has rating factor 1.
- Note that the different groups have different indicated rates.

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# Classification Ratemaking: Example

| Marital Status | Current Factor | Indicated Rate | Indicated Relativity |
|---|---|---|---|
| Single | 1.20 | $234 | 1.17 |
| **Married** | **1.00** | **$200** | **1.00** |
| Divorced | 1.10 | $222 | 1.11 |

- We calculate the indicated relativity as (Indicated Rate) / (Indicated Rate Base Class).
- We then make a selection for each class.

- Note that the indicated relativities need not average to 1.

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# Classification Ratemaking

- We use these new relativities in the formula

$$\text{Premium} = \text{Base Rate} \times \text{Factor1} \times \text{Factor2} \times \text{Factor3} \times \ldots$$

- We then adjust our base rate as needed, so the overall premium is increased by the indicated amount.
  - The **base rate** is the rate an individual in the base class for all rating variables would be charged.

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# Conclusion

- Pricing, both at the aggregate and individual level, is extremely intricate.
- We have touched only on the basics of the different steps involved.
- In practice, many refinements are made to every single step!

- (Not to mention, we have excluded estimation of expenses, which are extremely important as well!)

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